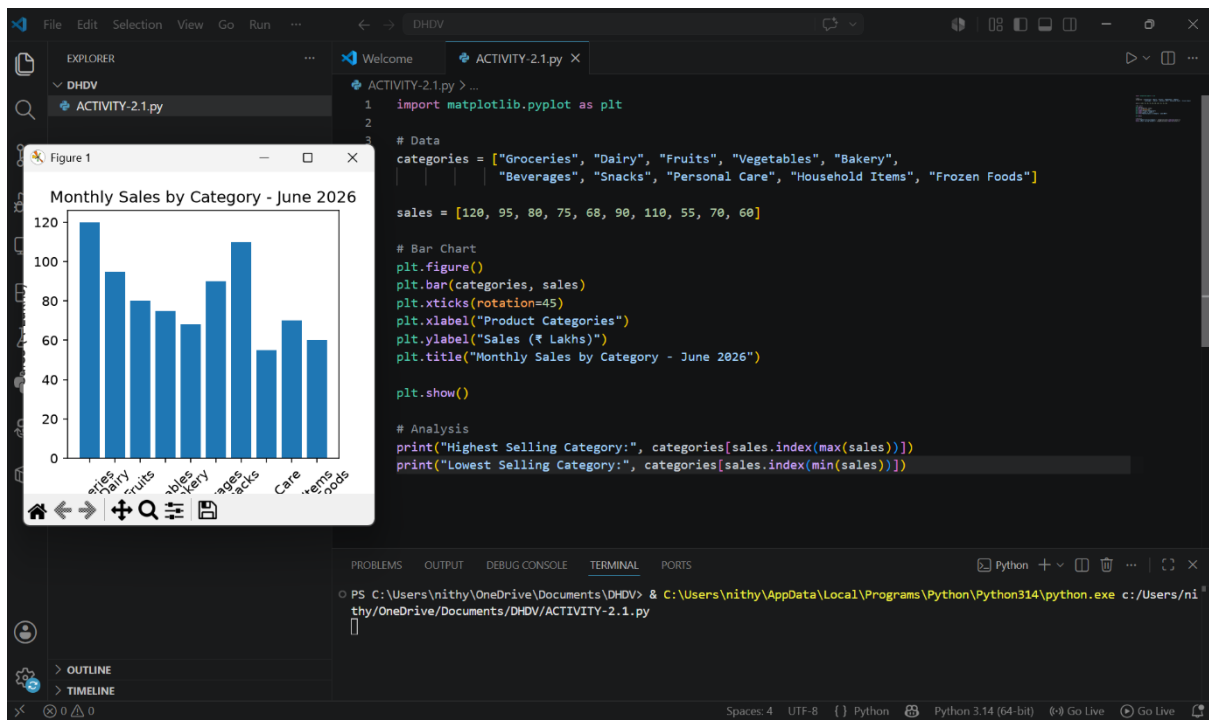


QUESTION-1



```
import matplotlib.pyplot as plt
```

```
# Data
```

```
categories = ["Groceries", "Dairy", "Fruits", "Vegetables", "Bakery",
              "Beverages", "Snacks", "Personal Care", "Household Items", "Frozen Foods"]
```

```
sales = [120, 95, 80, 75, 68, 90, 110, 55, 70, 60]
```

```
# Bar Chart
```

```
plt.figure()
```

```
plt.bar(categories, sales)
```

```
plt.xticks(rotation=45)
```

```
plt.xlabel("Product Categories")
```

```
plt.ylabel("Sales (₹ Lakhs)")
```

```
plt.title("Monthly Sales by Category - June 2026")
```

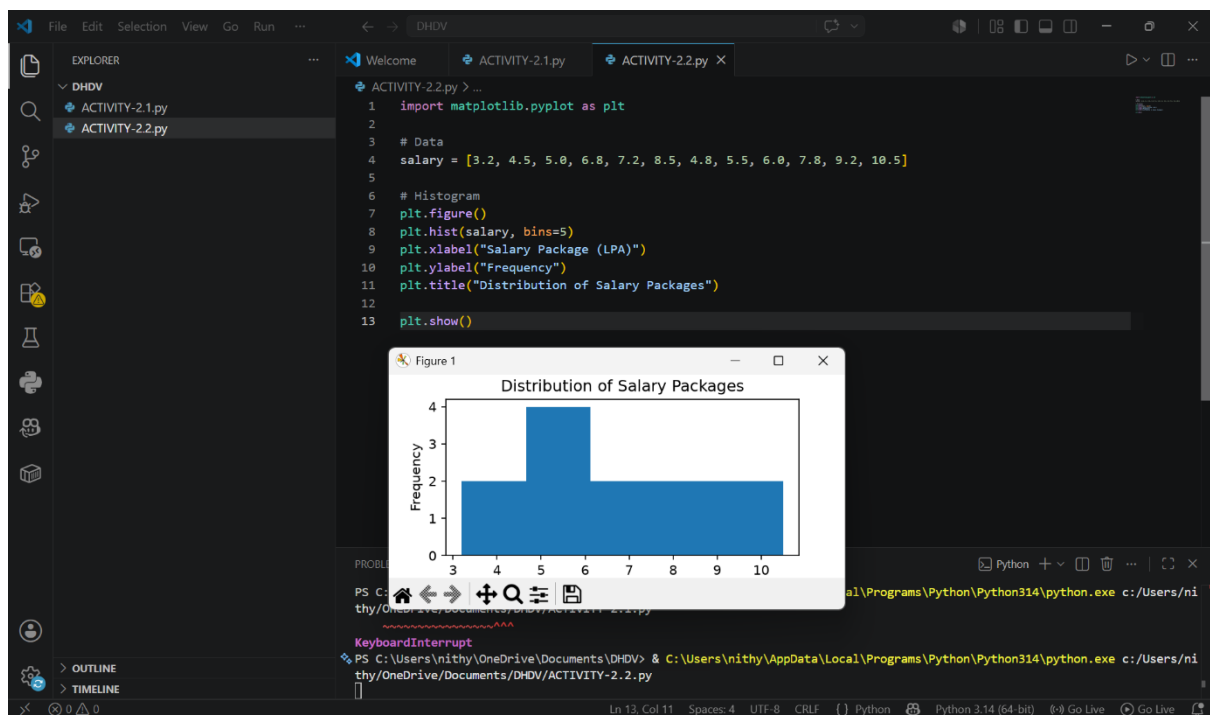
```
plt.show()
```

```
# Analysis
```

```
print("Highest Selling Category:", categories[sales.index(max(sales))])
```

```
print("Lowest Selling Category:", categories[sales.index(min(sales))])
```

QUESTION-2



```
import matplotlib.pyplot as plt
```

```
# Data
```

```
salary = [3.2, 4.5, 5.0, 6.8, 7.2, 8.5, 4.8, 5.5, 6.0, 7.8, 9.2, 10.5]
```

```
# Histogram
```

```
plt.figure()
```

```
plt.hist(salary, bins=5)
```

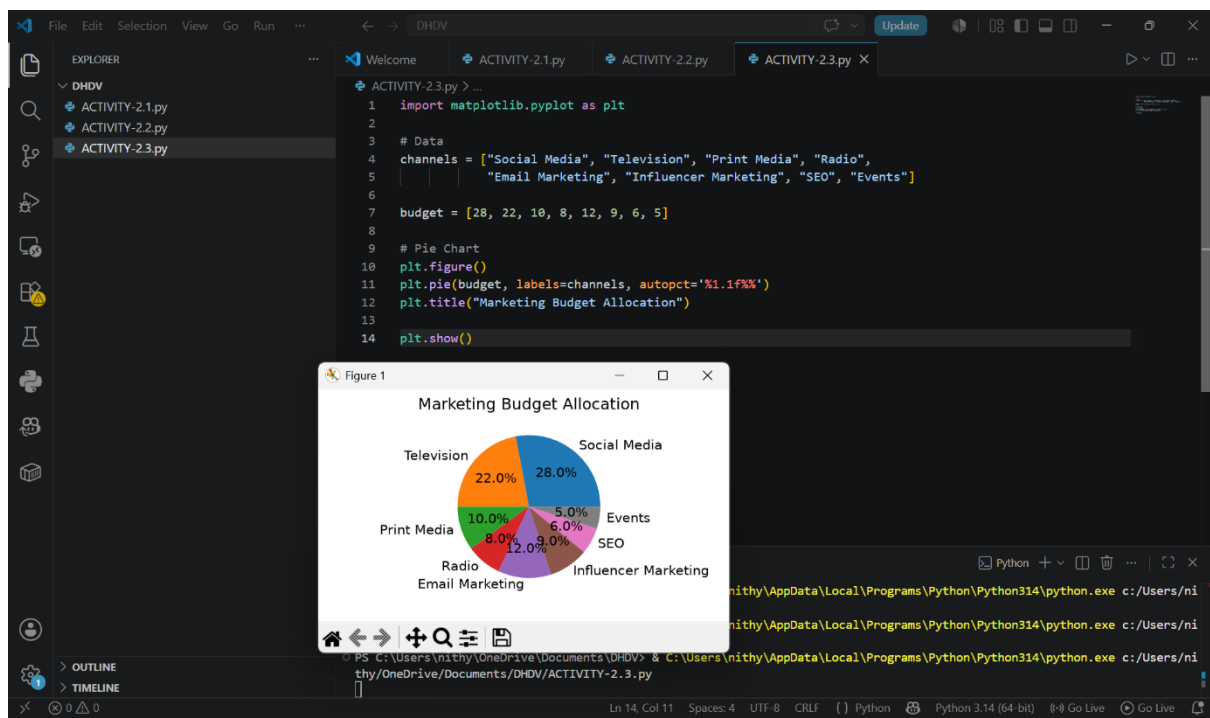
```
plt.xlabel("Salary Package (LPA)")
```

```
plt.ylabel("Frequency")
```

```
plt.title("Distribution of Salary Packages")
```

```
plt.show()
```

QUESTION-3



```
import matplotlib.pyplot as plt
```

```
# Data
```

```
channels = ["Social Media", "Television", "Print Media", "Radio",  
            "Email Marketing", "Influencer Marketing", "SEO", "Events"]
```

```
budget = [28, 22, 10, 8, 12, 9, 6, 5]
```

```
# Pie Chart
```

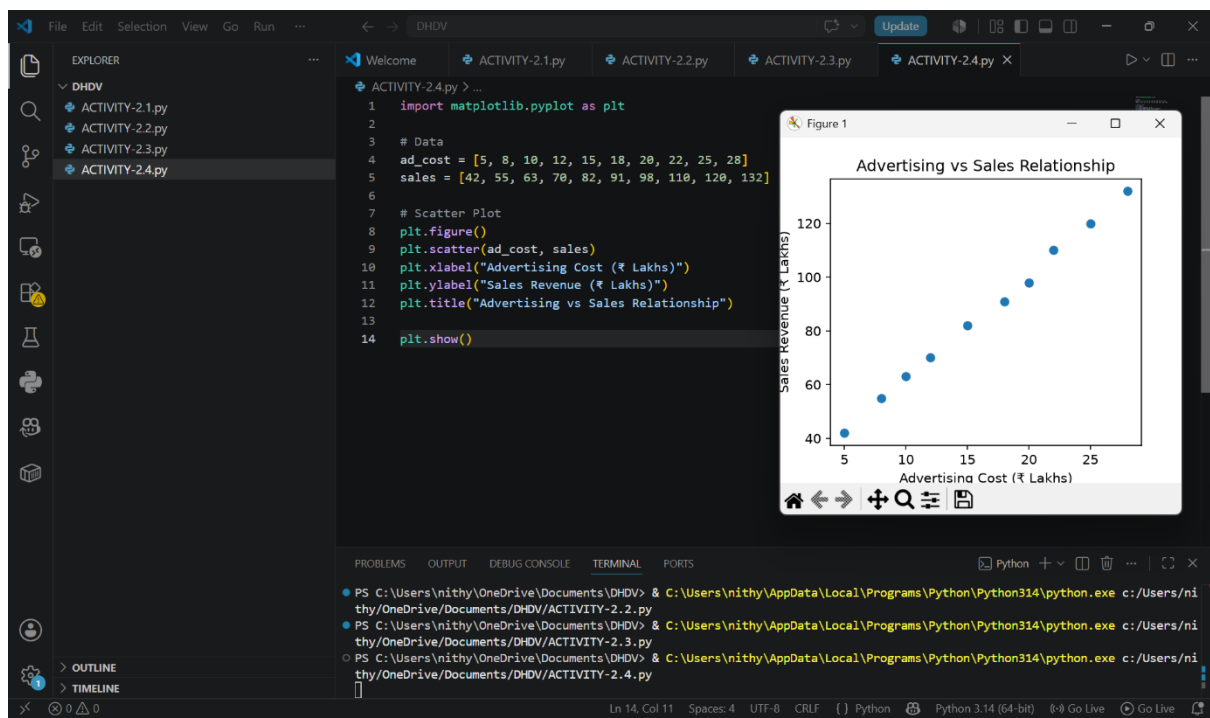
```
plt.figure()
```

```
plt.pie(budget, labels=channels, autopct='%1.1f%%')
```

```
plt.title("Marketing Budget Allocation")
```

```
plt.show()
```

QUESTION-4



```
import matplotlib.pyplot as plt
```

```
# Data
```

```
ad_cost = [5, 8, 10, 12, 15, 18, 20, 22, 25, 28]
```

```
sales = [42, 55, 63, 70, 82, 91, 98, 110, 120, 132]
```

```
# Scatter Plot
```

```
plt.figure()
```

```
plt.scatter(ad_cost, sales)
```

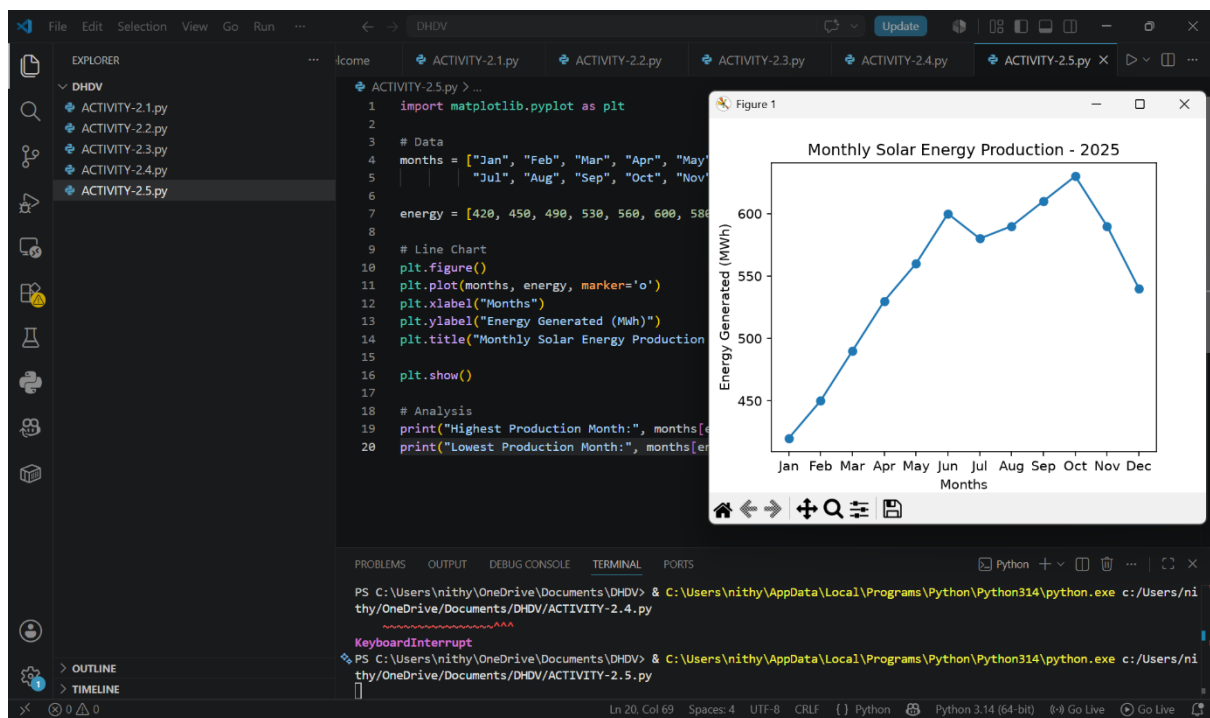
```
plt.xlabel("Advertising Cost (₹ Lakhs)")
```

```
plt.ylabel("Sales Revenue (₹ Lakhs)")
```

```
plt.title("Advertising vs Sales Relationship")
```

```
plt.show()
```

QUESTION-5:



```
import matplotlib.pyplot as plt
```

```
# Data
```

```
months = ["Jan", "Feb", "Mar", "Apr", "May", "Jun",
           "Jul", "Aug", "Sep", "Oct", "Nov", "Dec"]
```

```
energy = [420, 450, 490, 530, 560, 600, 580, 590, 610, 630, 590, 540]
```

```
# Line Chart
```

```
plt.figure()
```

```
plt.plot(months, energy, marker='o')
```

```
plt.xlabel("Months")
```

```
plt.ylabel("Energy Generated (MWh)")
```

```
plt.title("Monthly Solar Energy Production - 2025")
```

```
plt.show()
```

```
# Analysis
```

```
print("Highest Production Month:", months[energy.index(max(energy))])
```

```
print("Lowest Production Month:", months[energy.index(min(energy))])
```